

## R Code

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1
2
3 ##### Practical 1 #####
4
5 # Objectives:
6 #1 To understand the basic concepts of multivariate data and perform exploratory data
  analysis using R.
7 #2 To compute descriptive statistics, covariance matrix, and correlation matrix for a
  multivariate dataset.
8 #3 To visualize multivariate data using scatter plots, histograms, boxplots, scatter
  plot matrix, and correlation heatmap.
9
10
11 # Dataset
12 # We use the built-in iris dataset available in R.
13
14
15 # Packages Required
16
17 #1. corrplot
18 #2 GGally ( for plots)
19
20
21 ##### R code #####
22
23 install.packages("GGally")
24 install.packages("corrplot")
25
26 #####
27 # Load Packages
28 #####
29
30 library(GGally)
31 library(corrplot)
32
33 #####
34 # Load Dataset
35 #####
36
37 data(iris)
38
39 #####
40 # View Dataset
41 #####
42
43 head(iris)
44
45 tail(iris)
46
47 View(iris)
```

```
48
49 #####
50 # Structure of Dataset
51 #####
52
53 str(iris)
54
55 #####
56 # Dimensions
57 #####
58
59 dim(iris)
60
61 #####
62 # Variable Names
63 #####
64
65 names(iris)
66
67 #####
68 # Summary Statistics
69 #####
70
71 summary(iris)
72
73 #####
74 # Descriptive Statistics
75 #####
76
77 #####
78 # Mean
79 #####
80
81 colMeans(iris[,1:4])
82
83 #####
84 # Median
85 #####
86
87 apply(iris[,1:4],2,median)
88
89 #####
90 # Standard Deviation
91 #####
92
93 apply(iris[,1:4],2,sd)
94
95 #####
96 # Variance
97 #####
98
99 apply(iris[,1:4],2,var)
100
101 #####
```

```
102 # Minimum
103 #####
104
105 apply(iris[,1:4],2,min)
106
107 #####
108 # Maximum
109 #####
110
111 apply(iris[,1:4],2,max)
112
113 #####
114 # Covariance Matrix
115 #####
116
117 cov_matrix <- cov(iris[,1:4])
118
119 cov_matrix
120
121 #####
122 # Correlation Matrix
123 #####
124
125 cor_matrix <- cor(iris[,1:4])
126
127 cor_matrix
128
129 #####
130 # Scatter Plot
131 #####
132
133 plot(iris$Sepal.Length,
134       iris$Petal.Length,
135       col=iris$Species,
136       pch=19,
137       xlab="Sepal Length (cm)",
138       ylab="Petal Length (cm)",
139       main="Scatter Plot")
140
141 #####
142 # Scatter Plot Matrix
143 #####
144
145 pairs(iris[,1:4],
146        col=iris$Species,
147        pch=19,
148        main="Scatter Plot Matrix")
149
150 #####
151 # Enhanced Scatter Plot Matrix
152 #####
153
154 ggpairs(iris,
155          aes(color=Species))
```

```
156
157 #####
158 # Correlation Heatmap
159 #####
160
161 corrrplot(cor_matrix,
162           method="color",
163           type="upper",
164           addCoef.col="black",
165           tl.col="black",
166           tl.srt=45)
167
168 #####
169 # Histograms
170 #####
171
172 par(mfrow=c(2,2))
173
174 hist(iris$Sepal.Length,
175      col="skyblue",
176      main="Sepal Length",
177      xlab="Sepal Length")
178
179 hist(iris$Sepal.Width,
180      col="lightgreen",
181      main="Sepal Width",
182      xlab="Sepal Width")
183
184 hist(iris$Petal.Length,
185      col="pink",
186      main="Petal Length",
187      xlab="Petal Length")
188
189 hist(iris$Petal.Width,
190      col="orange",
191      main="Petal Width",
192      xlab="Petal Width")
193
194 #####
195 # Boxplots
196 #####
197
198 par(mfrow=c(1,1))
199
200 boxplot(iris[,1:4],
201        col="lightyellow",
202        main="Boxplots of Iris Variables",
203        ylab="Measurement (cm)")
204
205 #####
206 # Species-wise Boxplot
207 #####
208
209 boxplot(Sepal.Length~Species,
```

```
210     data=iris,
211     col=c("red","green","blue"),
212     main="Sepal Length by Species",
213     xlab="Species",
214     ylab="Sepal Length")
215
216 #####
217 # Species-wise Scatter Plot
218 #####
219
220 plot(iris$Petal.Length,
221      iris$Petal.Width,
222      col=iris$Species,
223      pch=19,
224      xlab="Petal Length",
225      ylab="Petal Width",
226      main="Petal Length vs Petal Width")
227
228 legend("topleft",
229       legend=levels(iris$Species),
230       col=1:3,
231       pch=19)
232
233 #####
234 # Pairwise Correlation Values
235 #####
236
237 round(cor_matrix,3)
238
239 #####
240 # Covariance Values
241 #####
242
243 round(cov_matrix,3)
244
245
246 ##### Practice Questions #####
247
248
249
250 #Using the built-in mtcars dataset, perform the following tasks:
251
252 #1 Load the dataset.
253 #2 Display the first six observations.
254 #3 Display the structure and dimensions of the dataset.
255 #4 Compute the mean, median, standard deviation, variance, minimum, and maximum for
    the numerical variables mpg, disp, hp, and wt.
256 #5 Compute the covariance matrix and correlation matrix.
257 #6 Draw:
258
259 #Scatter plot of Weight (wt) vs Mileage (mpg).
260 #Scatter plot matrix of the selected variables.
261 #Correlation heatmap.
262 #Histograms of the selected variables.
```

```
263 #Boxplots of the selected variables.  
264  
265 #7 Interpret the relationship between weight and fuel efficiency based on the scatter  
    plot and correlation matrix.  
266
```